

CHEN, Oscar

Hw2 Inference

1.

```
class_2_lecture.ipynb M x  arxiv_scraper.py 9  {} arxiv_clean.json x  Class 2 Homework.ipynb ●
hw2 > {} arxiv_clean.json > ...

1  [
2  {
3      "url": "https://arxiv.org/abs/2510.27688",
4      "paper_id": "2510.27688",
5      "title": "Continuous Autoregressive Language Models",
6      "abstract": "The efficiency of large language models (LLMs) is fundamentally limited by their sequential, token-by-token generation process.",
7      "authors": [
8          "Chenze Shao",
9          "Darren Li",
10         "Fandong Meng",
11         "Jie Zhou"
12     ],
13     "date": "[Submitted on 31 Oct 2025]",
14     "categories": "Computation and Language (cs.CL)",
15     "scraped_at": "2025-11-03T15:19:16.272228"
16 },
17 {
18     "url": "https://arxiv.org/abs/2510.27672",
19     "paper_id": "2510.27672",
20     "title": "Culture Cartography: Mapping the Landscape of Cultural Knowledge",
21     "abstract": "To serve global users safely and productively, LLMs need culture-specific knowledge that might not be learned from general-domain data.",
22     "authors": [
23         "Caleb Ziems",
24         "Yi Ma",
25         "Yi Ma",
26         "Yi Ma",
27         "Yi Ma",
28         "Yi Ma",
29         "Yi Ma",
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98         "Yi Ma",
99         "Yi Ma",
100        "Yi Ma"
101     ],
102     "date": "[Submitted on 31 Oct 2025]",
103     "categories": "Computation and Language (cs.CL)",
104     "scraped_at": "2025-11-03T15:19:16.272228"
105 }
106 ]
```

PROBLEMS 15 OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER GITLENS

✓ Scraping completed! Check arxiv_clean.json for all 2 papers.
PS C:\Users\kow12\OneDrive\Desktop\INFERENCE\hw2> python .\arxiv_scraper.py

=====

arXiv Paper Scraper
Category: cs.CL
OCR Available: True
=====

Starting arXiv scraper for category: cs.CL
Fetching papers from category: cs.CL
Fetching from: https://arxiv.org/list/cs.CL/recent?skip=0&show=50
Collected 50 paper URLs so far...
Fetching from: https://arxiv.org/list/cs.CL/recent?skip=50&show=50
Collected 100 paper URLs so far...
Fetching from: https://arxiv.org/list/cs.CL/recent?skip=100&show=50
Collected 150 paper URLs so far...
Fetching from: https://arxiv.org/list/cs.CL/recent?skip=150&show=50
Collected 200 paper URLs so far...

Found 200 papers to scrape

Scraping paper 1/200: https://arxiv.org/abs/2510.27688
Scraping paper 2/200: https://arxiv.org/abs/2510.27672
Scraping paper 3/200: https://arxiv.org/abs/2510.27641

2.

arxiv_pdf_ocr.ipynb X

hw2 > pdf_ocr > arxiv_pdf_ocr.ipynb > M+ arXiv PDF → Text (OCR) — Jupyter Notebook > M+ Quick Start > # If you don't have packages, uncomment the %pip lines below and run.

Generate + Code + Markdown | Run All Restart Clear All Outputs | Jupyter Variables Outline ...

arXiv PDF → Text (OCR) — Jupyter Notebook

Batch-download arXiv PDFs and extract text with **Tesseract OCR** while preserving a simple per-page structure.

This notebook was generated from your Python script and organized into runnable cells with a safer, OS-aware configuration.

Quick Start

1. **Set up dependencies** (one-time):
 - Install Poppler and Tesseract on your system.
 - Install Python libraries (see the next cell).
2. **Configure paths** (Windows users usually need to set `TESSERACT_PATH` and `POPPLER_PATH`).
3. **Provide `arxiv_clean.json`** (from your Task 1) in the same folder as this notebook.
4. **Run** the "Run batch OCR" cell.

```
# If you don't have packages, uncomment the %pip lines below and run.
# Internet might be required to install these.
# %pip install pdf2image pillow pytesseract requests

import sys, platform, os, json, time, re
from pathlib import Path
from datetime import datetime

import requests
from pdf2image import convert_from_path
import pytesseract
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER GITLENS

power

PS C:\Users\kow12\OneDrive\Desktop\INFERENCE>

transcription.py × Class 2 Homework.ipynb M

hw2 > task3 > transcription.py > ...

> path

```
1  """
2  Whisper Transcription Bot for YouTube Conference Talks
3  Downloads audio from YouTube videos and transcribes using Whisper AI
4  Performs OCR on video frames for text extraction
5  """
6
7  import os
8  import json
9  import time
10 from datetime import datetime
11 from pathlib import Path
12 import os, shutil
13
14
15 # ----- FFmpeg bootstrap (for Whisper) -----
16 FFMPEG_BIN = "C:\\ffmpeg\\bin" # change if you installed elsewhere
17 ffmpeg_exe = os.path.join(FFMPEG_BIN, "ffmpeg.exe")
18 ffprobe_exe = os.path.join(FFMPEG_BIN, "ffprobe.exe")
19
20 # add to PATH for this Python process
21 os.environ["PATH"] = FFMPEG_BIN + os.pathsep + os.environ.get("PATH", "")
22
23 # sanity print (should show full paths, not None)
24 print("ffmpeg:", shutil.which("ffmpeg"))
25 print("ffprobe:", shutil.which("ffprobe"))
26
27 if os.path.exists(ffmpeg_exe) and os.path.exists(ffprobe_exe):
28     os.environ["PATH"] = FFMPEG_BIN + os.pathsep + os.environ.get("PATH", "")
```

PROBLEMS 8 OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER GITLENS

ffmpeg: C:\\ffmpeg\\bin\\ffmpeg.EXE

✓ Successfully processed: A Short Introduction to Efficient Natural Language Processing #nlp

=====

BATCH PROCESSING COMPLETE

=====

Successfully processed: 10/10 videos

Total time: 19.8 minutes

Output location: transcriptions\\transcripts

=====

✓ Summary saved to: transcriptions\\talks_transcripts.json

=====

✓ Transcription completed!

✓ Check the 'transcriptions/transcripts/' folder

=====

o (.venv) PS C:\\Users\\kow12\\OneDrive\\Desktop\\INFERENCE\\hw2\\task3> |

3.

4.

